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LAB ASSIGNMENT 02

Using Wireshark to Investigate the Transmission of Dummy Login Credentials over HTTP and HTTPS

OBJECTIVE:

To analyze the transmission of dummy login credentials over HTTP and HTTPS using Wireshark and compare the captured packets to demonstrate how HTTPS encryption prevents sensitive information from being intercepted and read by unauthorized users.

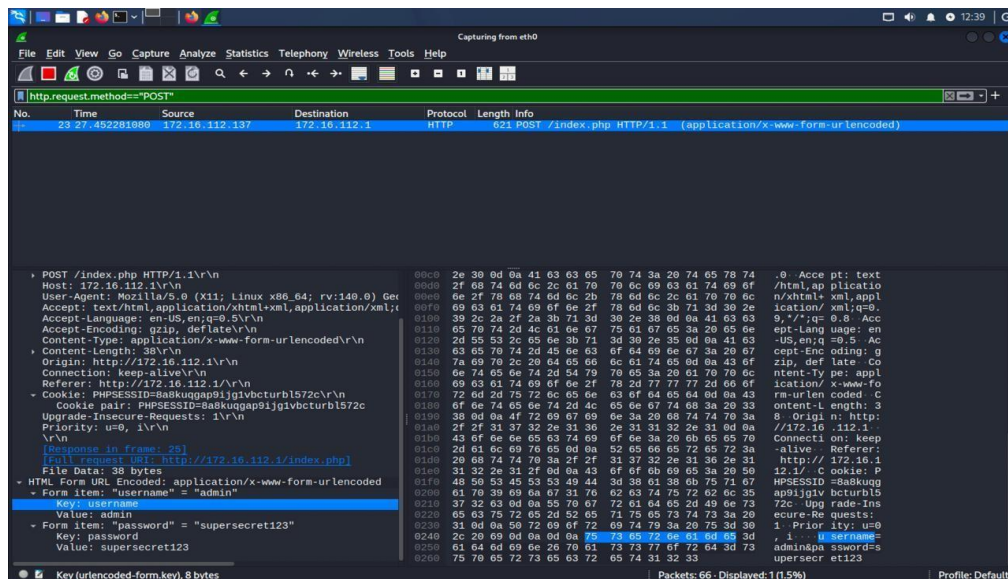
TOOLS:

- Wireshark
- Web browser
- HTTP test login website
- GitHub login page

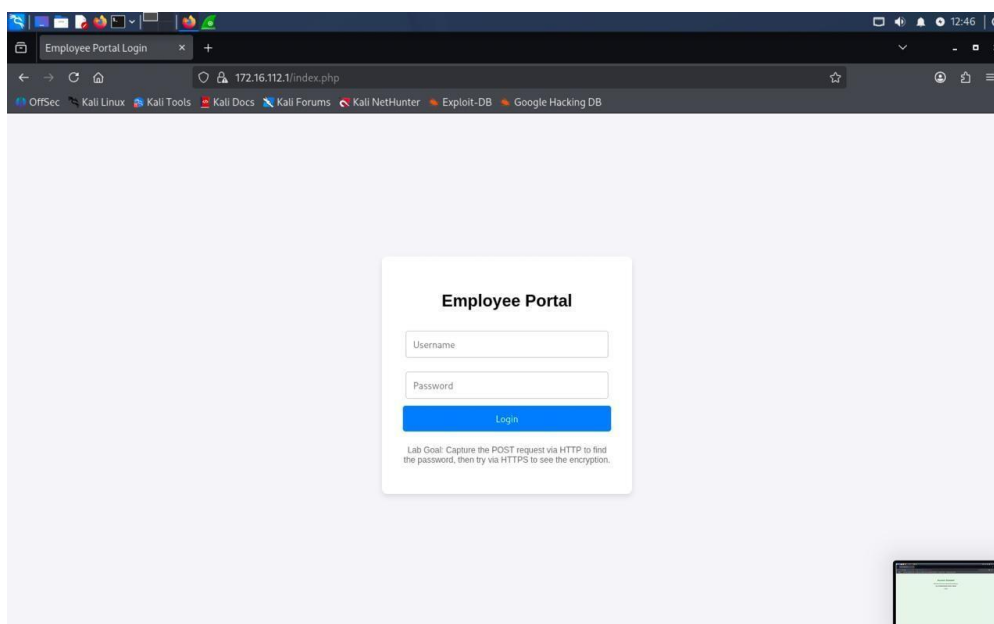
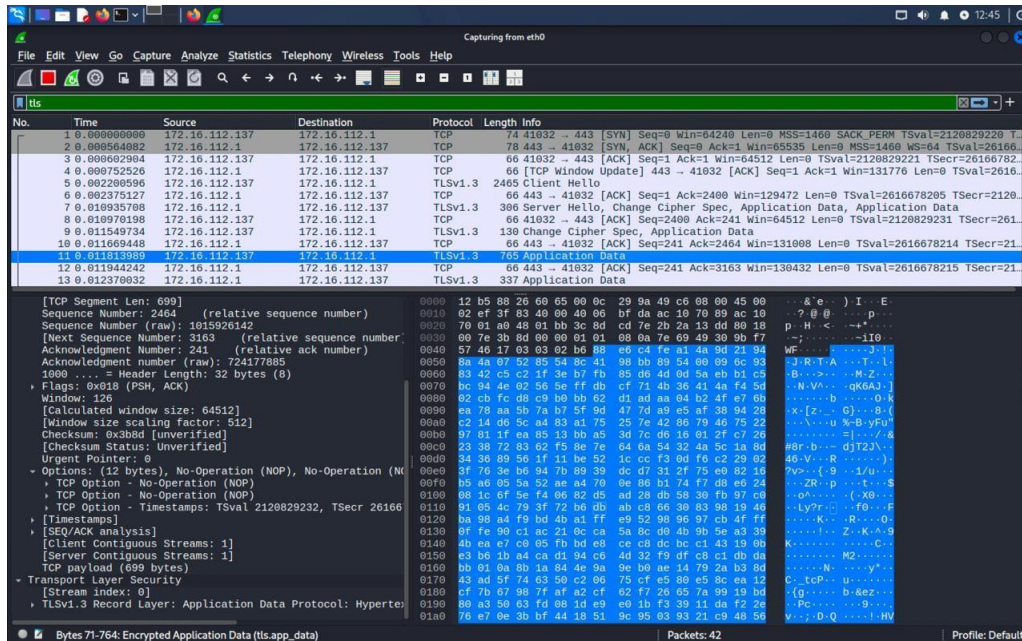
Procedure

Network traffic was captured with Wireshark during two separate login experiments: the first used an HTTP-based test login page, while the second used an HTTPS-protected login page.

HTTP Login: A packet capture was initiated on the active eth0 network interface before accessing the HTTP test login website. Dummy credentials were then entered and submitted. After the submission, the capture was stopped and the HTTP POST request was inspected. The username and password could be viewed directly in readable form within the submitted form data.



HTTPS Login: A fresh packet capture was started on the same network interface before accessing the GitHub login page, which communicates using HTTPS. Dummy login credentials were entered and submitted, and the capture was then stopped for analysis. In contrast to HTTP, the credentials could not be viewed in readable form because TLS encryption protects the transmitted data.



Comparison

Features	HTTP	HTTPS
Data Transmission	Data is transmitted without encryption, so login information may be visible in Wireshark.	Data is encrypted using TLS, so the actual login information cannot be directly read.
Security & Confidentiality	Does not provide confidentiality, making sensitive information vulnerable to interception	Provides confidentiality and better protection for sensitive information during transmission.
Use for Login	It should not be used to securely transmit usernames and passwords.	Recommended for login pages and other communications involving sensitive information.

Importance of Cryptography

Cryptography plays an important role in protecting sensitive information during network communication. In this experiment, the dummy credentials transmitted through HTTP were visible in Wireshark because the data was sent without encryption. When HTTPS was used, TLS encrypted the transmitted information, making the credentials unreadable in the captured packets. The experiment therefore demonstrates how cryptographic protection reduces the risk of sensitive data being exposed to someone monitoring network traffic and supports the confidentiality and integrity of information during transmission.

Conclusion

This experiment demonstrated the practical security difference between HTTP and HTTPS. In the HTTP packet capture, the dummy login credentials were visible in Wireshark, whereas the HTTPS capture contained encrypted data that could not be directly interpreted as the submitted credentials. The results clearly illustrate how encryption helps safeguard sensitive information, such as usernames and passwords, during network transmission.